



Smart Garbage Reporting & Analytics System

A Digital Civic Complaint & Municipal Analytics Platform — built for smarter, cleaner, and more transparent cities.



SMART CITY INITIATIVE

</> OPEN SOURCE

The Problem: Broken Civic Systems

Cities are growing faster than their sanitation infrastructure. Citizens have no effective way to report issues — and municipalities lack the tools to respond.

Slow Complaint Systems

Manual processes cause delays and lost reports with no accountability trail.

No Real-Time Monitoring

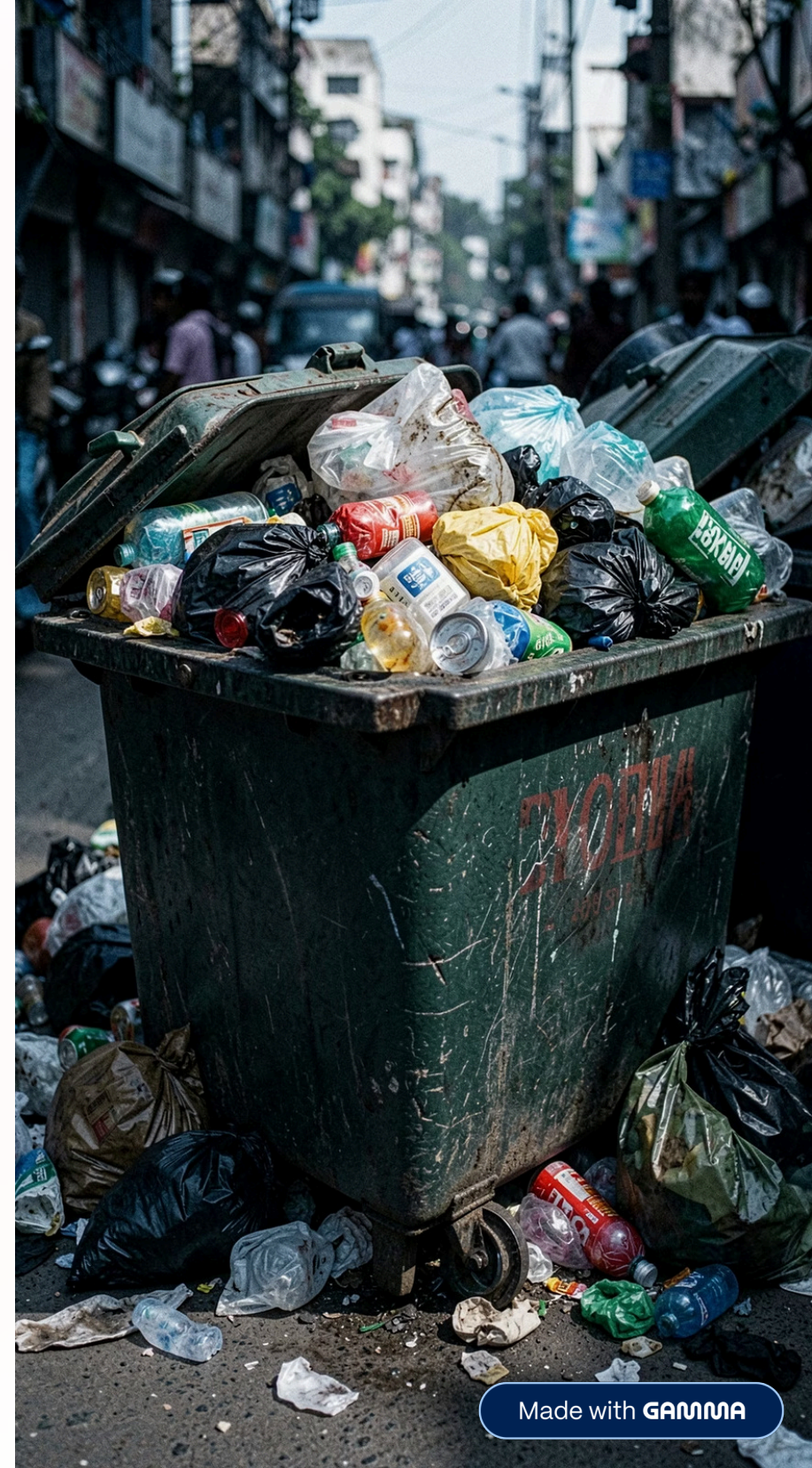
Municipalities operate blind — no live visibility into sanitation status.

Lack of Transparency

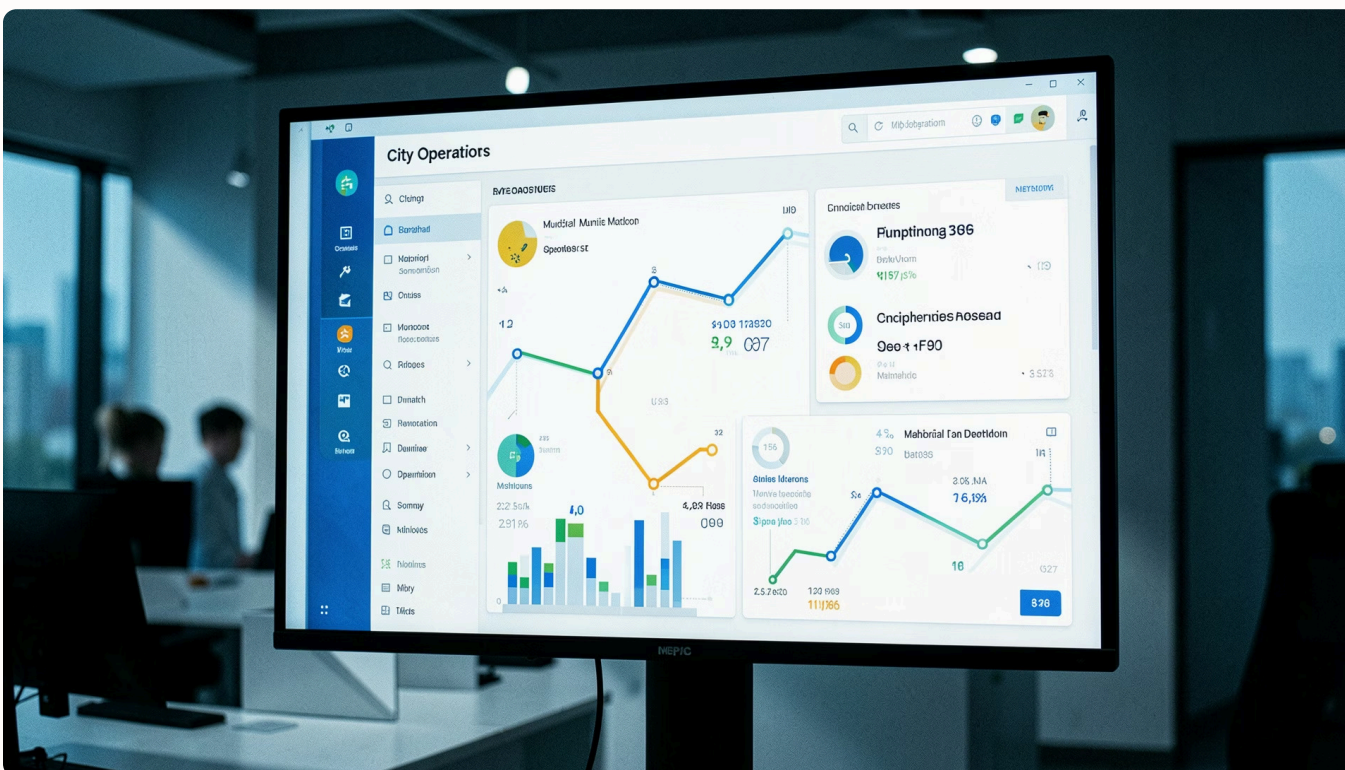
Citizens never know if their complaint was received, assigned, or resolved.

Poor Citizen Engagement

Low participation due to friction in reporting and zero feedback loops.



Our Solution: A Unified Civic Platform



The **Smart Garbage Reporting & Analytics System** bridges the gap between citizens and municipal authorities — enabling instant reporting, real-time tracking, and data-driven decision-making.

- **Citizen Reporting Portal**
Submit complaints with GPS location and photo evidence in seconds.
- **Donation & Fund Support**
Integrated QR, UPI, and Razorpay for cleanliness fund contributions.

- **Admin Dashboard**
Live analytics, complaint management, and fund tracking in one view.
- **Multi-Language Support**
Accessible to all citizens regardless of language preference.

Core Platform Features

Every feature is designed to reduce friction for citizens and empower administrators with actionable intelligence.



Complaint Submission

Simple form with image upload and live GPS location sharing.



Live Location Tracking

Auto-capture GPS coordinates for precise issue mapping.



Photo Evidence Upload

Citizens attach images to strengthen complaint validity.



Admin Authentication

Secure login with session management and access control.



Analytics Dashboard

Real-time charts, status breakdowns, and trend monitoring.



Donation & Payment

QR, UPI, and Razorpay integration for cleanliness fund support.



Multi-Language Support

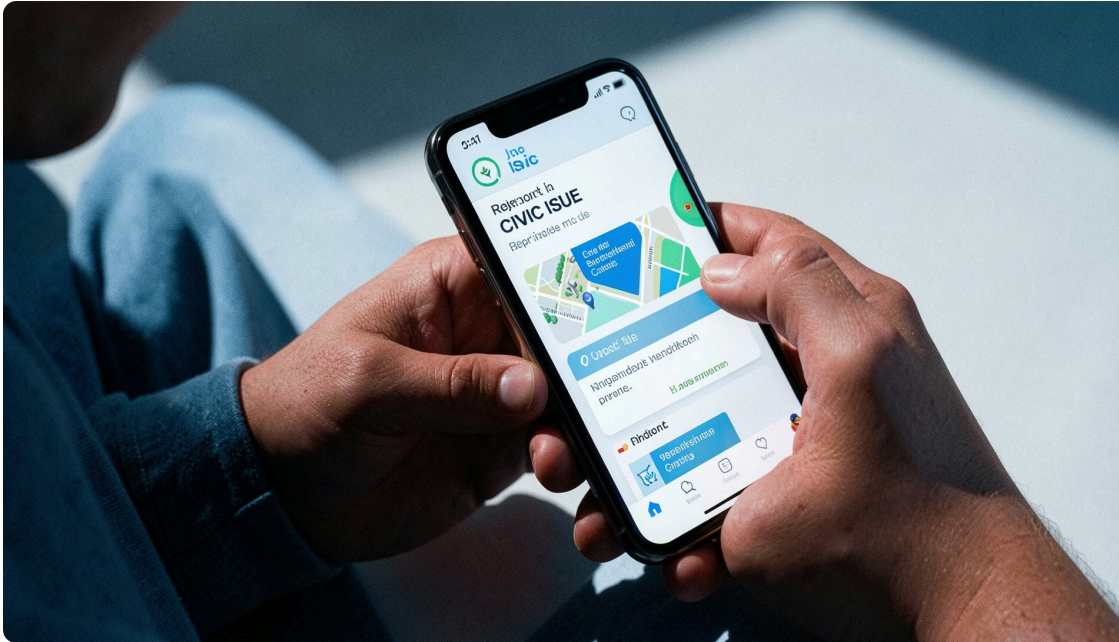
Localized interface ensuring accessibility across diverse populations.



Social Media Integration

Share complaints and updates directly to social platforms.

Citizen Portal Experience



Frictionless Reporting for Every Citizen

The citizen-facing portal is designed for speed and simplicity — anyone can file a complaint in under 60 seconds.

Complaint Form

Category selection, description, and priority tagging.

GPS Auto-Capture

Location shared automatically — no manual input needed.

Status Tracking

Live updates on complaint progress from submission to resolution.

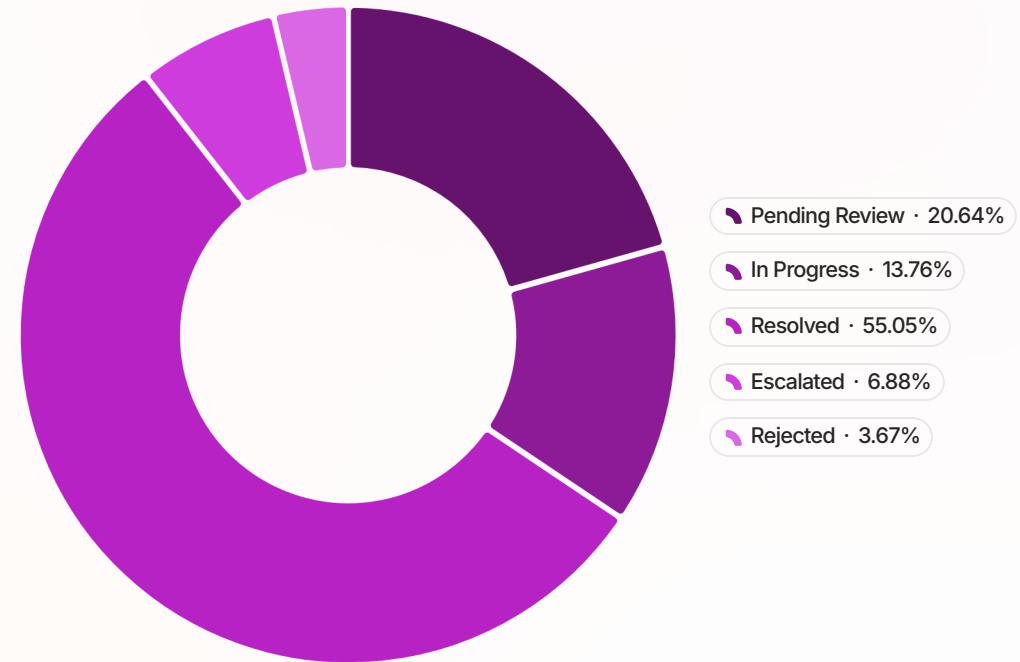
Donation Module

Contribute to city cleanliness funds via QR or UPI.

Admin Dashboard & Analytics

Full Operational Visibility

Municipal administrators gain a command-center view of all sanitation activity — enabling faster decisions and better resource allocation.



Pending vs. Resolved

Live ratio tracking of complaint status.

Fund Usage Reports

Transparent tracking of donations and expenditures.

Social Media Feed

Integrated civic communication management.

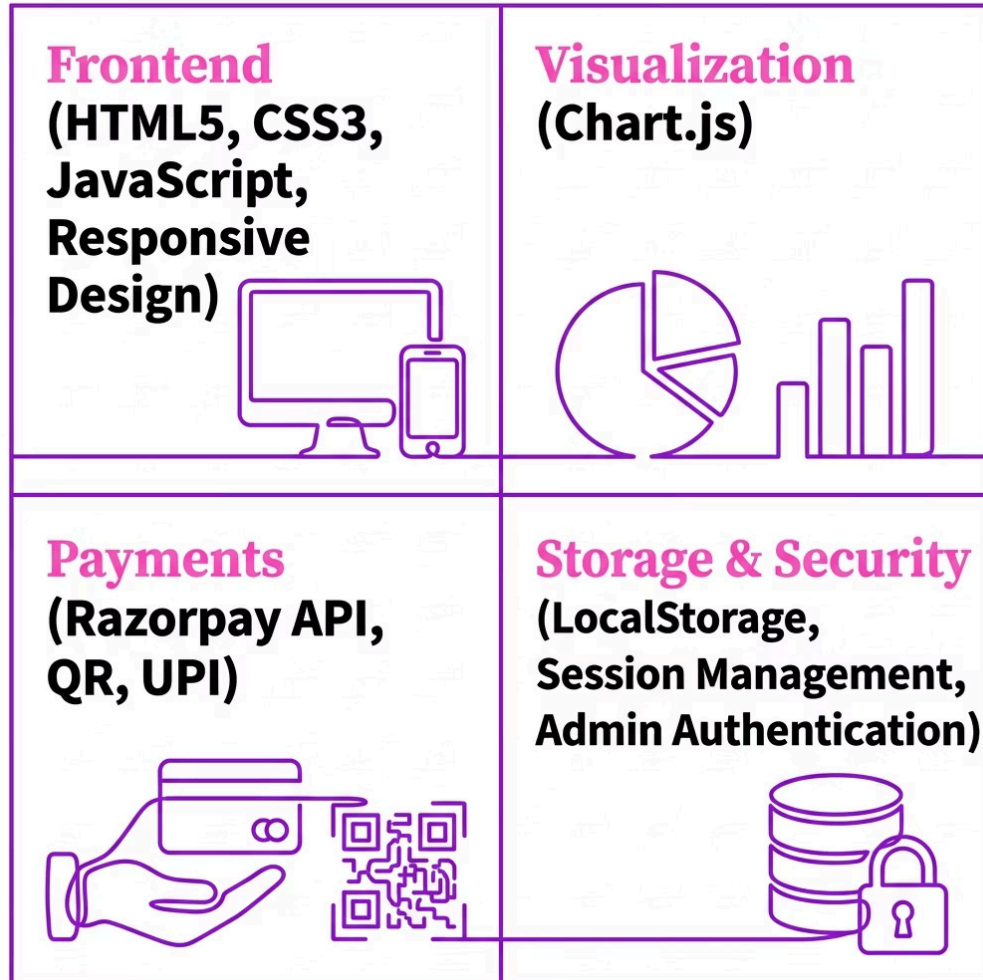
System Architecture & Workflow

A streamlined, end-to-end pipeline ensures every complaint is captured, routed, acted upon, and tracked — with zero manual handoff delays.



The architecture is lightweight and browser-based — leveraging **LocalStorage** for data persistence and **Chart.js** for real-time visualization, making it deployable without complex server infrastructure.

Technology Stack



Built for Simplicity & Scale

The platform uses a modern, lightweight stack — no heavy backend required. Fully responsive across all devices and screen sizes.

HTML5 + CSS3

Glassmorphism UI with clean, modern card-based design.

JavaScript + Chart.js

Dynamic analytics and interactive data visualizations.

Razorpay API

Secure, compliant payment processing for donations.

LocalStorage

Offline-capable data persistence without server dependency.

Security, Advantages & Future Roadmap

Security First

Admin authentication, session management, secure Razorpay integration, and browser storage protection ensure data integrity at every layer.

Key Advantages

Smart governance, faster response times, full transparency, improved cleanliness, active citizen participation, and alignment with Digital India initiatives.

What's Next

AI-powered waste detection, IoT smart dustbins, native mobile app, GIS mapping, real-time municipal alerts, and cloud database integration.



Thank You

The **Smart Garbage Reporting & Analytics System** is more than a project — it's a blueprint for smarter, cleaner, and more accountable cities. Built for scale.

Designed for impact.

 **GitHub Repository**

[View Source Code](#)

 **Live Demo**

Explore the platform and submit a test complaint.

 **Q&A**

Open floor for questions, feedback, and collaboration.

 **Smart City Ready.** Scalable, open-source, and built for real-world municipal deployment.

